Assessing How External Data Address Information Gaps in Montandon-Based Disaster Preparedness Question Answering

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***Abstract*—Disaster preparedness and the development of Early Action Protocols (EAPs) require evidence about past hazards, impacts, response activities, and financing. This evidence is distributed across structured disaster records, news reports, weather observations, and spatial data. We investigate which information needs remain unanswered in Montandon-based disaster question answering and whether external sources help address those gaps.**

We construct a Neo4j graph that connects Montandon disaster and impact records with IFRC GO response information, news articles, NASA POWER weather data, and disaster footprints. A language-model assistant uses predefined graph-query tools to retrieve evidence and answer questions. For country–hazard contexts, we compare answers produced with the core graph against answers produced when one external source is enabled. The comparison examines both the answers and the evidence retrieved for five EAP-related question areas.

***Index Terms*—Disaster preparedness, information gaps, knowledge graphs, question answering, external data, Early Action Protocols.**

I. Introduction

Disaster preparedness requires more than knowing when and where a hazard occurred. Planners also need evidence about affected places and people, past response activities, participating organizations, and available financing. An Early Action Protocol (EAP) specifies prearranged actions and resources for acting ahead of a forecast hazard. Historical disaster and response records can inform several parts of that planning process [1].

The Montandon Global Crisis Data Bank brings together structured records from multiple disaster-related collections. Nevertheless, a question relevant to preparedness may require information that is not fully expressed in a single event or impact record. News reports may describe local conditions and response activities; weather data can provide environmental context; and spatial footprints can describe the extent or intensity of a hazard. Whether those additional sources answer a particular question is an empirical issue.

This paper asks which information needs remain unanswered in Montandon-based disaster question answering and how selected external sources help address them. We build a graph-based question-answering system around Montandon records and compare answers to the same country–hazard question under different source-access conditions. The analysis focuses on information relevant to EAP development, including hazard context, vulnerability, coordination, early actions, and financing.

The contributions of this work are threefold. First, we construct a graph representation that connects disaster events with impacts, response records, news articles, weather summaries, and spatial footprints while retaining source information. Second, we provide a question-answering interface based on predefined, validated graph queries. Third, we define a paired evaluation procedure for examining which requested information is supplied, qualified, corroborated, or left unresolved when an external source is made available.

II. Related Work

Structured disaster databases support the study of hazard occurrence and reported impacts. EM-DAT documents disaster events and their effects [2]. The Geocoded Disasters (GDIS) dataset extends EM-DAT with subnational spatial information, showing how location data can help connect disaster records to additional sources [3]. Our work similarly uses linked sources, but focuses on their contribution to answers to preparedness-related questions.

Text sources can provide information beyond structured fields. Li et al. used language models and other natural-language-processing methods to extract records of climate-extreme impacts from online text [4]. In our system, news articles are linked to disaster events and made available as retrieved evidence. We evaluate whether access to this evidence changes the answer to a specific question.

Graph-based retrieval offers another way to organize evidence for language models. GraphRAG builds a graph from documents and uses summaries of related entities to answer broad questions [5]. Our graph begins with structured disaster records and explicit source relationships. The assistant selects among predefined Cypher-query tools instead of generating arbitrary Cypher queries.

III. Data Sources

A. Montandon Disaster Records

Montandon records form the central disaster evidence in the system. The collections used for graph construction include event and impact records from EM-DAT, GDACS, and the IFRC event collection. Event records provide identifiers, titles, descriptions, dates, countries, and hazard codes. Impact records provide reported impact types, categories, values, units, and estimate types. Record identifiers and correlation identifiers are retained during preparation so that related records can be linked in the graph.

B. IFRC GO Response Records

IFRC GO provides response-event and appeal records. Response events describe emergencies and associated country information. Appeals provide information such as amounts requested and funded and the number of people targeted. Their source event identifiers are retained so that an appeal can be linked to its corresponding IFRC GO event.

C. News

News candidates are located through Media Cloud. Article metadata includes titles, publication dates, outlets, and URLs. Where accessible, article bodies are retrieved from the linked publisher pages. Candidates are classified according to whether they directly cover the specified disaster event, provide related background, concern another event, or remain uncertain. Articles classified as directly covering the event are linked to that event in the graph.

D. Weather

NASA POWER supplies daily weather data for locations associated with Montandon records. The pipeline processes temperature, precipitation, and wind measurements and produces summaries for the associated records. Stored summaries include values, units, the retrieval period, and counts of days with available measurements.

E. Spatial Footprints

Spatial footprints describe a hazard's geographic extent or intensity. The system processes Copernicus Emergency Management Service (CEMS) mapping layers and GDACS polygons available through Montandon. CEMS layers may describe observed hazard extent, flood extent or depth, and mapped buildings, transport, or facilities. GDACS polygons provide source-defined hazard areas.

For qualifying Montandon earthquake records, the system additionally retrieves USGS ShakeMap data. ShakeMap intensity values are grouped into spatial bands at Modified Mercalli Intensity IV or above. These bands describe modeled shaking intensity, rather than a confirmed inventory of damage. Each footprint retains its source, layer, characterization, summary attributes, and links to the underlying spatial data.

IV. System and Methods

A. System Overview

The workflow has four stages: source-specific data preparation, graph construction, evidence retrieval and question answering, and source-contribution evaluation. Prepared records and source snapshots are stored in S3. The graph-building pipeline validates the prepared inputs and loads their graph representations into Neo4j. Fig. 1 shows how the sources move through these stages.

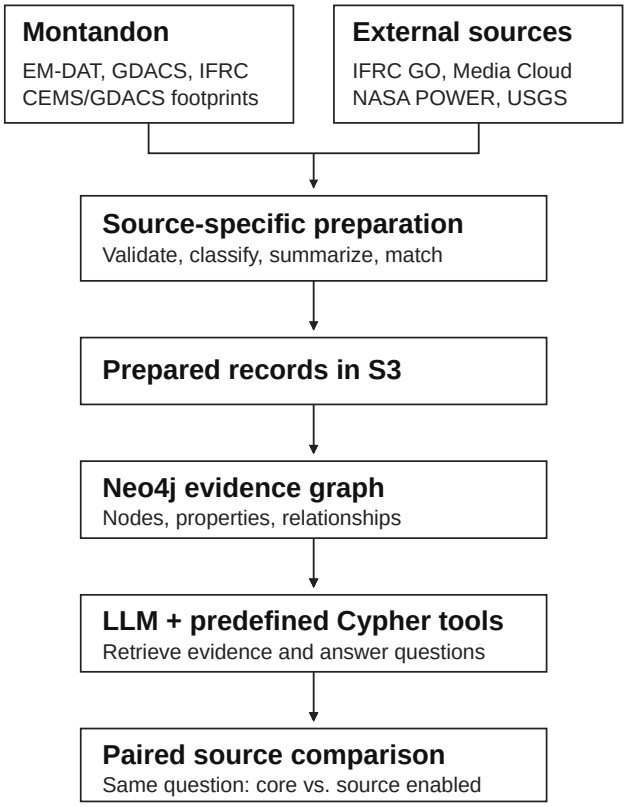


Fig. 1. System workflow from Montandon and external sources to graph-based answers and paired source comparison.

B. Neo4j Graph Model

Montandon records are represented as `MontandonItem` nodes and receive `Event` or `Impact` labels according to their record roles. An `IMPACT_OF` relationship connects impacts to events with the same correlation identifier. Events are linked to country, hazard, and start-date nodes. Description, keyword, and impact-related embeddings support bounded semantic-similarity relationships between distinct Montandon records.

IFRC GO events and appeals are represented as `GOEvent` and `GOAppeal` nodes. GO events connect to country nodes, while `FOR_EVENT` links each appeal to the GO event identified in its source record. A shared country node provides geographic context; it does not establish that a GO event and a Montandon event are the same disaster.

Fig. 2 distinguishes graph relationships from the identifier match used to attach weather properties.

C. Connecting Sources to Disaster Records

The sources require different linkage procedures. News articles do not share a Montandon event identifier. The news pipeline therefore compares each candidate with a target event using location, timing, hazard, and reported effects. An article classified as covering the same event becomes a `NewsArticle` node linked to the target `Event` by `VERIFIED_FOR`. This relationship records an event-relevance decision; it does not validate every factual claim in the article.

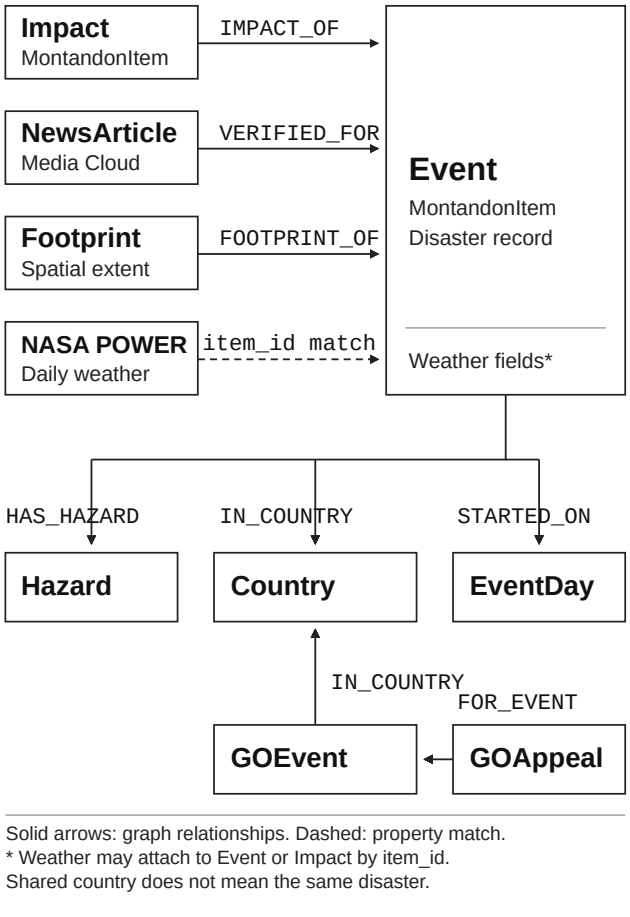


Fig. 2. Graph links around a disaster event. Solid arrows are relationships; the dashed arrow indicates the identifier match used for weather properties.

Weather summaries are attached to the existing `MontandonItem` whose identifier matches the collected weather result's `item_id`. Weather is represented as node properties rather than a separate weather node.

Footprints are linked to `Event` nodes using matching correlation identifiers and `FOOTPRINT_OF` relationships. The graph stores footprint summaries and source URLs rather than full GeoJSON geometry. Retrieved event context distinguishes Montandon-origin footprints from external USGS ShakeMap footprints.

D. Evidence Retrieval and Answer Generation

A language-model assistant selects from predefined graph-query tools for disaster-event searches, event-context retrieval, related-event searches, IFRC GO response and appeal searches, and news retrieval. Tool arguments are validated before a predefined Cypher query is executed. The assistant uses the returned records to compose an answer and may make further calls within a configured call budget.

Weather summaries and footprint information are returned with disaster context when available. News articles are retrieved through a separate `get_event_news` tool. This tool returns saved articles linked to a selected event, including metadata and article excerpts. The assistant is instructed to retrieve relevant missing evidence before concluding that it is unavailable, but tool selection remains model-driven. Consequently, evaluation records the tool calls and returned evidence as well as the final answer.

V. Evaluation Design

A. EAP-Related Questions

Evaluation contexts are defined by country and hazard. Five fixed question templates are instantiated for each selected context: (1) hazard context and timing, (2) affected areas and vulnerable groups, (3) responding organizations and coordination, (4) past actions and needs relevant to early-action choices, and (5) appeal funding and people targeted. These questions concern historical evidence that may inform EAP development.

B. Paired Source Comparisons

For each question, the system first generates a core answer with news, NASA POWER weather, and external ShakeMap footprints disabled. The core condition retains Montandon records, Montandon-origin footprints, and access to IFRC GO tools. The same core answer is compared separately with an answer generated after enabling one selected external source: news, weather, or external ShakeMap footprints.

Source access is controlled at the tool or returned-evidence level. Disabling news removes the news-retrieval tool. Disabling weather removes weather fields and weather-based search filters. Disabling external footprints removes the external-footprint fields while preserving Montandon-origin footprints. Comparisons therefore measure the effect of making a source available within this question-answering workflow. Tool-call records are examined to determine which evidence the assistant actually retrieved.

C. Contribution Categories

A judge compares the two answers, their retrieved evidence, and the question. The rubric assigns one of seven categories: `fills_field` when external evidence supplies a requested value missing from an existing core field; `extends_coverage` when it answers a requested aspect outside the core fields; `qualifies` when it materially corrects or limits an existing answer; `corroborates` when it supports an existing answer without changing it; `core_sufficient` when core evidence already answers the question; `unresolved` when requested information remains missing; and `unsupported` when an important new or changed claim is not supported by retrieved evidence.

The rubric checks for unsupported claims first. Additional detail counts as an improvement only when it addresses the question and is supported by relevant evidence. For each source and question type, the improvement rate is the number of evaluated comparisons labeled `fills_field`, `extends_coverage`, or `qualifies`, divided by the number of evaluated comparisons. Answer and judgment failures are recorded separately.

VI. Experimental Setup

VII. Results

VIII. Discussion

IX. Conclusion

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